

Quantification: 682_z29

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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
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lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2
gap bridges added (detection)	17

Interior lacunae only: mean, median and sample SD (n = 11)

measure	kind	mean	median	sd	units
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area_px2	lacuna shape	1740.55	1624	939.642	px^2
major_axis_length_px	lacuna shape	75.1181	75.9278	23.0542	px
minor_axis_length_px	lacuna shape	29.0371	30.0551	9.2359	px
aspect_ratio	lacuna shape	2.7064	2.5263	0.7296	unitless
eccentricity	lacuna shape	0.9096	0.9183	0.0622	unitless
solidity	lacuna shape	0.87	0.8778	0.0425	unitless
roots_count	headline	5.5455	5	1.6348	count
ring_length_r30_px	headline	250.273	247	97.6044	px
ring_length_r60_px	reported	697.455	702	302.281	px
owned_length_px	ownership-dependent	848.515	799.784	602.715	px
edge_count	ownership-dependent	26.1818	22	22.1667	count
mean_edge_length_px	ownership-dependent	38.2007	36.5334	13.1105	px
perimeter_px	normalised	201.309	214.167	60.6733	px
ring_area_r30_px2	normalised	7938.55	8595	2078.18	px^2
ring_area_r60_px2	normalised	19815	20090	4945.82	px^2
in_frame_fraction_r30	normalised	0.9943	1	0.0188	unitless
in_frame_fraction_r60	normalised	0.981	1	0.0583	unitless
ring_density_r30	normalised	0.0315278	0.0310652	0.0098891	px^-1
ring_density_r60	normalised	0.0345586	0.0335953	0.0101788	px^-1
roots_per_100px_perimeter	normalised	2.8431	2.7484	0.7371	per 100 px
ring_attached_length_r30_px	network v2	177.273	174	70.7758	px
ring_attached_length_r60_px	network v2	349.364	288	159.24	px
ring_length_w_r30_px	network v2	277.084	272.309	106.876	px
ring_length_w_r60_px	network v2	775.437	767.257	337.603	px
sholl_crossings_r10	network v2	6.2727	6	2.149	count
sholl_crossings_r20	network v2	9.4545	10	3.7514	count
sholl_crossings_r30	network v2	11.4545	11	4.5247	count
ring_width_mean_r30_px	width	5.0813	5.0395	0.4771	px

Per field

measure	value	units
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analysed_area_px2	1.02357e+06	px^2
total_skeleton_length_px	38368	px
canalicular_length_density_per_px	0.0374846	px^-1
junction_count	485	count
junction_density_per_px2	0.00047383	px^-2
lacuna_count	13	count
lacunae_per_px2	1.27e-05	px^-2

skeleton_component_count	707	count
mean_component_length_px	54.2687	px
median_component_length_px	16	px
field_length_density_w_per_px	0.0419867	px^-1
field_density_without_flagged_per_px	0.0368831	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.7908	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	8	px
width_pixels_used	34940	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
10	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
11	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
12	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
13	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	3	187	411	102.669	3	34.223	
2	8	375	1100	1038.8	35	29.6801	5.6666
3	4	309	954	1414.54	61	23.1892	5.0395
4	8	352	1197	2207.05	71	31.0853	5.3045
5	5	240	702	799.784	22	36.3538	4.8698
6	6	352	768	840.269	23	36.5334	5.202
7	9	319	716	857.169	37	23.1667	5.0452
8	5	247	662	1252.46	32	39.1394	4.3423
9	5	143	391	410.635	11	37.3304	4.6182
10	4	139	377	405.433	10	40.5434	6.0582
11	3	97	352	213.794	3	71.2646	4.8201
12	7	313	795	490.09	10	49.009	4.8817
13	6	186	374	260.794	10	26.0794	5.0914

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0333512	0.0297417	1.1446
2	229.522	9151	23871	1	0.999	0.0409791	0.046081	3.4855
3	120.811	6134	17911	1	1	0.050375	0.0532634	3.311
4	309.279	11331	28206	1	1	0.0310652	0.0424378	2.5867
5	235.238	9387	21888	1	1	0.0255673	0.0320724	2.1255
6	214.167	8637	20090	1	1	0.0407549	0.038228	2.8016
7	322.836	9940	22198	0.8402	0.7724	0.0320926	0.0322552	2.7878
8	229.48	8595	22042	1	1	0.0287376	0.0300336	2.1788
9	109.498	4245	11066	1	0.9865	0.0336867	0.0353335	4.5663
10	161.817	5491	13376	1	1	0.0253142	0.0281848	2.4719
11	147.196	6914	19480	1	1	0.0140295	0.0180698	2.0381
12	254.693	9728	23664	1	1	0.0321752	0.0335953	2.7484
13	202.693	7711	16371	0.9376	0.8055	0.0241214	0.0228453	2.9601

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	62	92	209.924	456.576	4	8	6
2	223	425	407.321	1247.89	9	12	19
3	235	440	343.522	1052.52	8	10	11
4	278	677	390.321	1326.45	10	13	18
5	133	278	260.338	767.257	5	12	11
6	283	523	399.463	865.767	7	13	13
7	289	516	349.037	792.37	10	11	11
8	174	288	272.309	727.671	7	10	9
9	124	286	160.61	434.848	5	5	6
10	123	251	159.953	416.463	4	5	7
11	72	138	110.882	402.961	3	4	7
12	193	373	341.522	875.826	6	14	16
13	112	164	201.681	412.161	5	6	9

